

Deadlock Handling Strategies

Meghna AG(38)

Deadlock handling is highly complicated in distributed system because of:

- Limited visibility:

No site have a complete picture of the entire system's state. Each site can only see the resources and processes it manages directly.

- Unreliable communication:

Communication between sites is slow, unreliable and have finite and unpredictable delay.

Three strategies for handling deadlock:

1. Deadlock prevention:
 - i. Process acquires all needed resources simultaneously before executing.
 - ii. Preempting a process that holds the needed resource.
2. Deadlock avoidance:

Resource is granted to a process only if the resulting global system state is safe(no possibility of a deadlock occurring).

Deadlock detection and avoidance is inefficient and impractical in distributed system.

3. Deadlock detection:

Requires the examination of the status of process-resource interactions for the presence of cyclic wait.

Deadlock detection is the best approach to handle deadlocks in a distributed system.

Consider some processes and resources.

A **wait-for** relationship exists between a process and a resource if the process is waiting to acquire that resource in order to proceed.

A **cyclic wait** occurs when there's a chain of wait-for relationships that loop back on itself.

